LENANA SCHOOL 2025 PREDICTION

POSSIBLE KCSE EXAM

BIOLOGY

231/2

PAPER 2

TIME: 2 HOURS

NAME………………………………………………………………………………………

SCHOOL……………………………………………… SIGN………………………

INDEX NO…………………………………………… ADM NO……………………

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES:

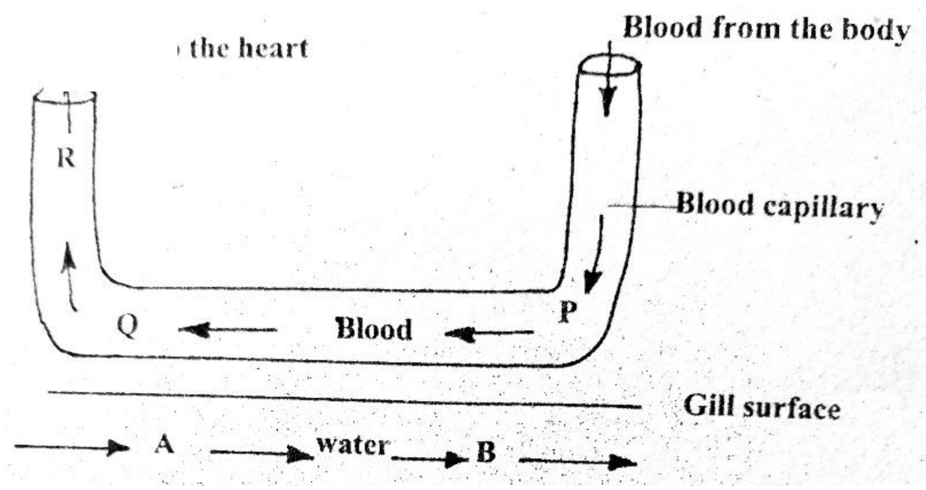
- *Write your **name** and **admission number** in the spaces provided.*
- *Answer **all** the questions in this paper in the spaces provided.*
- ***Answer questions 1-6 (compulsory) and either question 7 or 8.***

FOR EXAMINER'S USE ONLY:

QUESTIONS	MAXIMUM SCORE	CANDIDATE'S SCORE
1	8	
2	8	
3	8	
4	8	
5	8	
6	20	
7 or 8	20	

*Answer **all** the questions in this paper in the spaces provided.*

1.The diagram below represents the direction of flow of blood in a gill capillary. The percentage of oxygen in solution at position A, B, P, Q and R is given in the table below.



Position	Oxygen concentration in solution (%)	Haemoglobin saturation with oxygen (%)
A	10	
B	7	
P	4	55
Q	7	85

a) Why is the oxygen percentage low at P? **(1mk)**

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b) Using evidence from the data given, suggest what will happen to oxygen in the water at point B. **(3mks)**

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c) Name the organ into which blood coming from the capillary at Q flows.

(1mk)

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d) Suppose the flow of blood in the capillary illustrated above was in the opposite direction, explain the disadvantage it would have to the fish. **(2mks)**

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e) Name the principle where the blood flows in the opposite direction to another fluid. **(1mk)**

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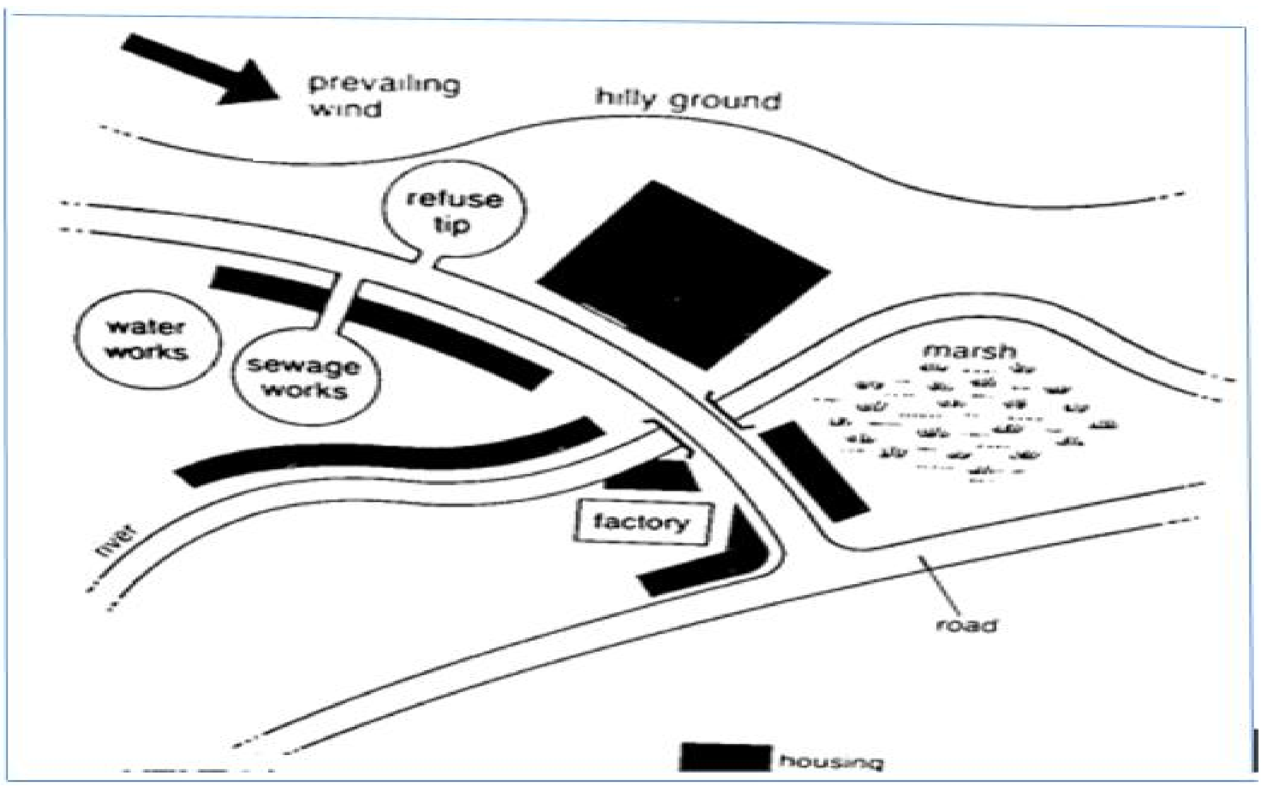
2. Below is a diagram of a poorly planned town showing some building and facilities.



a) Giving evidence from the diagram, state two likely sources of water pollution. **(2mks)**

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b) State three ways that the positioning of the refuse pit and sewage works pose danger to the residence of the town. **(3mks)**

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c) Residents living close to the marsh are likely to suffer from malaria. Explain. **(1mk)**

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d) Suggest two control measures to overcome water pollution in the area. **(2mks)**

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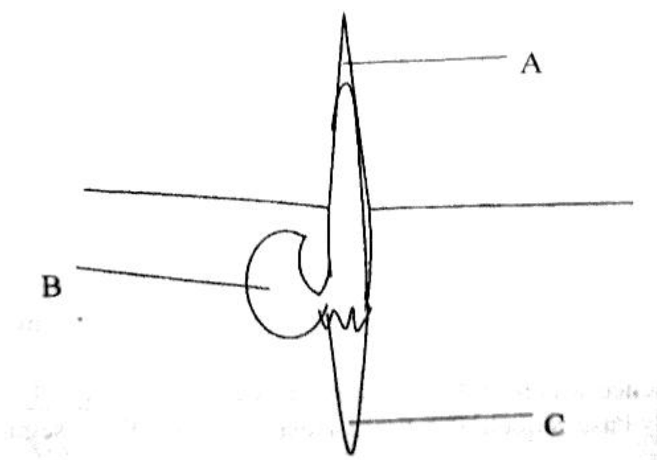
3. The diagram below represents a maize seedling.



a) Name the structure labeled A and C

(2mks)

A --

C ---

b) State the functions of parts labeled A, B and C**.**

(3mks)

A ---

B ---

C---

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c) Name the type of germination exhibited by maize

(1mk)

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d) Name two conditions necessary for seed germination other than water and oxygen. **(1mk)**

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e) What is the role of oxygen in seed germination?

(1mk)

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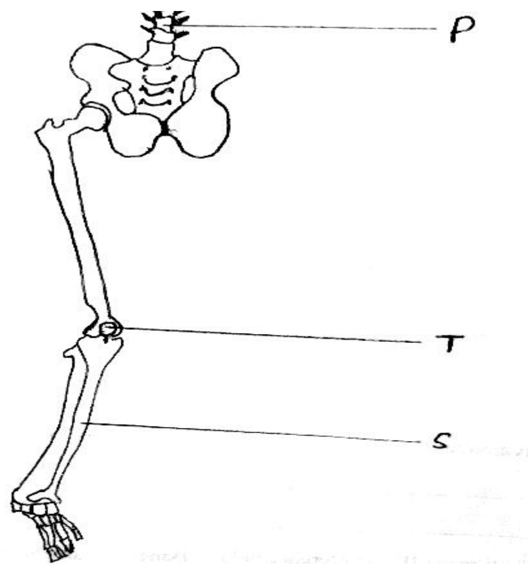
4. The figure below shows part of a human skeleton.



a) Which part of the human skeleton is it**?**

(1mk)

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b) On the diagram label by name three types of joints**.**

(3mks)

c) Label the S, T and P.

(3mks)

S ---

T ---

P ---

d) Which two bones on the diagram manufactures red blood cells?

(1mk)

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5. In maize the gene for purple colour is dominant to the gene for white colour.

A pure breeding maize plant with purple grains was crossed with a heterozygous plant.

a) Using letter G to represent the gene for purple colour, work out the genotypes of the offspring.

(4mks)

b) State the phenotype of the offspring**.**

(1mk)

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c) What is genetic engineering?

(1mk)

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d) Gene for smooth seed coat is dominant over gene for wrinkled seed coat. Two heterozygous pea plants with smooth seed coats were crossed and produced a total of 14640 seeds. How many seeds had wrinkled seed coat? Show your calculations. **(2mks)**

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6. The diagram below is obtained from measurements of growth in the leaf petiole of a certain plant. The relative growth rate is calculated and the data is obtained as shown below.

Time in days	0	1	2	3	4	5	6	7	8	9
Relative growth rate(cm/day)	0	0.1	0.3	0.8	2.0	4.0	4.5	3.5	0.2	0

a) Plot a graph of relative growth rate against time.

(5mks)

b) State two functions of a leaf petiole. **(2mks)**

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c) State two characteristics of cells found in the region of cell division.

(2mks)

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d) Account for the shape of the curve between the following **days**

(3mks)

i) 2 – 5.

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ii) 6 – 8 **(3mks)**

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iii)6 – 8

(3mks)

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d) Distinguish between primary growth and secondary growth in a flowering plant. **(2mks)**

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7. How are flowers adapted to wind and insect pollination? **(20mks)**

8a) Name factors that affects the enzyme controlled reactions. **(6mks)**

b) Explain the factors that affect the rate of enzyme activity. **(14mks)**

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